

Module4Lab1

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1 Module 4 Lab 1: Describing Data

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Course: CS 82A

Repo: <https://github.com/raulmckay-blip/cs82a-portfolio>

1.1 Overview

This lab uses the sales data I cleaned in Module 3 (`sales_clean.csv`) to practice describing data: center, spread, shape, and how two columns move together.

1.2 Step 1: Load the clean data

My file comes out as (292, 7) instead of (292, 6) because I kept the `is_return` column from the Module 3 cleaning.

```
[ ]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df = pd.read_csv('sales_clean.csv')
print(df.shape)    # (292, 7)
df.head()
```

(292, 7)

```
[ ]:  order_id    date  product  price  qty  zip  is_return
0      1254  2026-04-06      mug   11.36   4  60614      False
1      1057  2026-05-30  charger   14.79   3  30303      False
2      1150  2026-05-06      mug    5.50   1  10001      False
3      1066  2026-05-30  notebook  37.53   1  98101      False
4      1114  2026-04-06  webcam   45.59   4  90405      False
```

1.3 Step 2: Center and spread of price and qty

```
[ ]: for col in ['price', 'qty']:
      print(f"{col:>5} mean={df[col].mean():.2f} median={df[col].median():.2f}  ↳
      ↳std={df[col].std():.2f}")
```

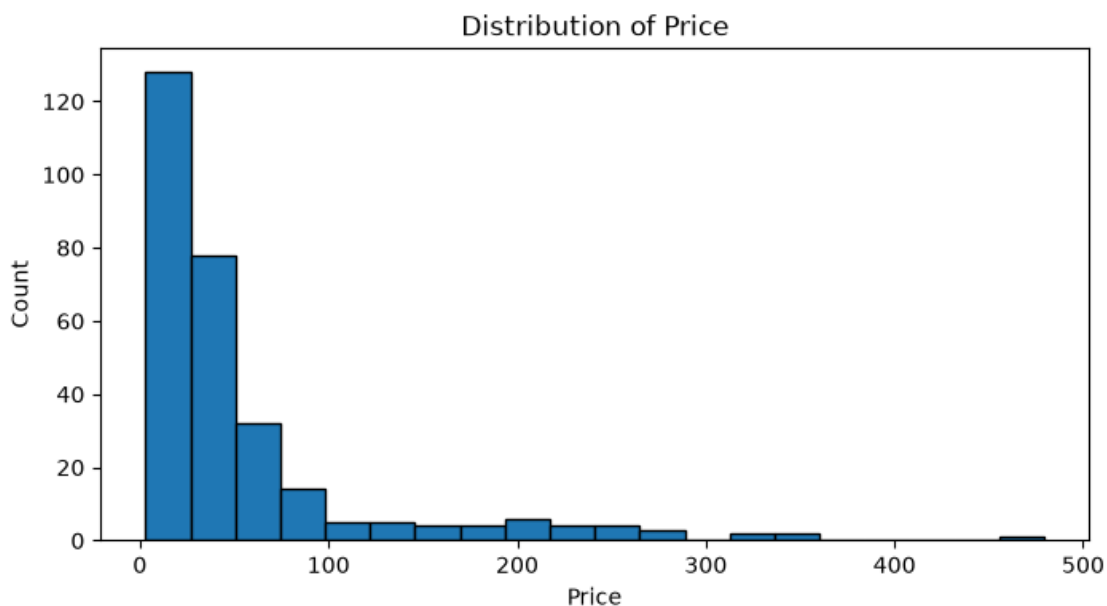
```
price mean=55.52 median=37.53 std=71.77
qty mean=2.91 median=3.00 std=1.56
```

1.4 Step 3: Which summary would I report for price?

I would report the median. The mean is \$55.52 but the median is only \$37.53, so a few expensive items are pulling the mean up. The median gives a better idea of what a normal item actually costs.

1.5 Step 4: Histogram of price

```
[ ]: plt.figure(figsize=(8, 4))
      plt.hist(df['price'], bins=20, edgecolor='black')
      plt.title('Distribution of Price')
      plt.xlabel('Price')
      plt.ylabel('Count')
      plt.show()
```



The histogram is right skewed. Most prices are bunched up on the low end and there's a long tail of expensive items to the right.

This matches what I said in Step 3. Skew pulls the mean toward the tail, which is why the median is the better choice here.

1.6 Step 5: Create the revenue column

```
[ ]: df['revenue'] = df['price'] * df['qty']
      print(df.shape)    # should be 8 columns now
      df.head()
```

(292, 8)

```
[ ]:   order_id      date  product  price  qty   zip  is_return  revenue
0      1254  2026-04-06      mug   11.36   4  60614      False    45.44
1      1057  2026-05-30  charger   14.79   3  30303      False    44.37
2      1150  2026-05-06      mug    5.50   1  10001      False     5.50
3      1066  2026-05-30  notebook   37.53   1  98101      False    37.53
4      1114  2026-04-06   webcam   45.59   4  90405      False   182.36
```

1.7 Step 6: Correlation matrix

I left out `order_id` and `zip`. They're stored as numbers but they're really just labels, so a correlation on them wouldn't mean anything.

```
[ ]: corr = df[['price', 'qty', 'revenue']].corr()
      corr.round(2)
```

```
[ ]:   price  qty  revenue
price    1.00 -0.02    0.82
qty     -0.02  1.00    0.33
revenue  0.82  0.33    1.00
```

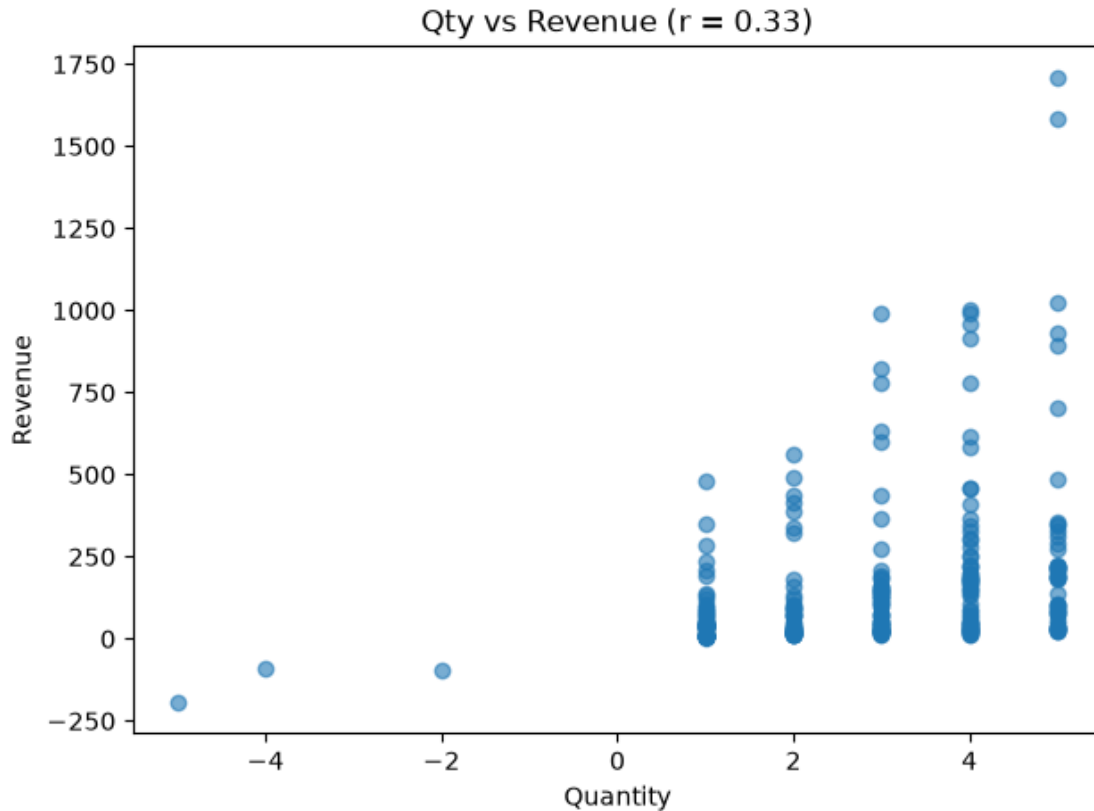
```
[ ]: # find the strongest pair (skipping the 1s on the diagonal)
      pairs = corr.where(~np.eye(len(corr), dtype=bool)).abs().unstack().dropna().
      ↪sort_values(ascending=False)
      top = pairs.index[0]
      print(f"Strongest pair: {top[0]} vs {top[1]}  r = {corr.loc[top[:2]].round(2)}")
```

Strongest pair: price vs revenue r = 0.82

The strongest pair is price vs revenue at $r = 0.82$. The lab said qty vs revenue should be the standout, but in my data it's only 0.33. I think it's because price varies way more than qty (std of 71.77 vs 1.56), so price ends up driving most of the difference in revenue.

1.8 Step 7: Scatter plot to verify

```
[ ]: r = df['qty'].corr(df['revenue'])
      plt.figure(figsize=(7, 5))
      plt.scatter(df['qty'], df['revenue'], alpha=0.6)
      plt.title(f'Qty vs Revenue (r = {r:.2f})')
      plt.xlabel('Quantity')
      plt.ylabel('Revenue')
      plt.show()
```



The points drift upward but it's loose. There's a lot of spread at each qty level, which makes sense for an r of 0.33.

1.9 Step 8: The analyst's sentence

Orders with more units tend to bring in more revenue, but it's only a moderate relationship ($r = 0.33$), not a tight one. One thing to be careful about is that price is really skewed, so a few expensive items can swing revenue a lot more than qty does.

1.10 Step 9: By-hand check

Data: [2, 4, 4, 6, 9]

Mean: $2 + 4 + 4 + 6 + 9 = 25$, and $25 / 5 = 5.0$

Median: The list is already sorted and has 5 numbers, so the median is the 3rd one, which is 4.

```
[ ]: vals = [2, 4, 4, 6, 9]
      print('mean  :', np.mean(vals))
      print('median:', np.median(vals))
```

```
mean  : 5.0
median: 4.0
```

1.11 AI Use

I used AI to help with syntax and refine wording on this lab.